

IRVINE AGCM, AB, Canada

WMO: 717950

Lat: 49.9892N

Lon: 110.2614W

Elev: 2608

StdP: 13.36

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.1	-14.6	-26.4	1.4	-19.9	-20.3	2.0	-14.5	30.8	31.3	27.8	28.8	6.0	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.9	91.0	61.3	87.4	61.2	83.5	60.4	65.5	80.4	63.8	79.7	62.3	78.8	11.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	87.0	71.1	58.4	80.4	68.4	56.5	74.9	66.9	31.8	80.1	30.4	79.6	29.2	78.8	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.0	25.3	22.4	DB	-29.1	97.1	4.1	3.0	-32.0	99.3	-34.4	101.0	-36.7	102.7	-39.7	104.8	(4)
(5)			WB	-29.3	69.2	4.0	2.2	-32.2	70.8	-34.5	72.1	-36.8	73.4	-39.7	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	19.5	17.0	31.0	42.4	52.9	60.8	67.3	66.4	57.3	43.5	30.8	17.8		
	DBStd	21.44	15.80	15.72	14.65	9.26	8.30	6.50	5.80	6.96	9.31	10.12	13.82	16.03		
	HDD50	4653	945	924	594	257	64	4	0	0	32	250	584	999		
	HDD65	8555	1410	1343	1053	677	385	158	40	67	265	667	1026	1464		
	CDD50	1868	0	0	5	30	153	327	537	509	251	49	7	0		
	CDD65	298	0	0	0	0	9	31	113	110	34	1	0	0		
	CDH74	4420	0	0	0	14	182	475	1623	1566	535	25	0	0		
	CDH80	1629	0	0	0	3	31	142	621	625	204	3	0	0		
	WSAvg	10.3	10.8	9.6	9.5	11.4	9.7	10.6	9.4	8.8	9.2	11.8	11.1	11.3		
	PrecAvg	15.4	0.8	0.6	0.8	1.4	2.2	3.1	1.5	1.3	1.3	0.9	0.7	0.7		
(15) Precipitation	PrecMax	23.3	1.5	1.5	1.5	2.6	3.7	6.8	5.7	3.5	2.5	2.3	1.7	1.5		
	PrecMin	9.1	0.2	0.1	0.1	0.4	0.4	0.7	0.2	0.3	0.0	0.2	0.2	0.1		
	PrecStd	2.8	0.3	0.4	0.4	0.6	0.9	1.5	1.1	0.8	0.7	0.5	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	53.1	65.1	74.7	83.9	90.8	95.3	95.0	91.8	77.5	67.4	53.5		
		MCWB	40.5	42.8	47.6	52.7	57.7	61.6	63.0	62.2	59.7	56.5	49.1	42.0		
	2%	DB	46.9	45.7	59.6	69.5	79.1	83.6	90.9	90.7	86.3	69.8	58.7	47.8		
		MCWB	39.0	37.7	45.0	50.1	56.4	60.8	62.3	61.2	59.2	52.1	45.8	38.9		
	5%	DB	43.5	41.4	54.5	64.1	74.8	79.1	86.7	87.2	80.2	64.8	53.7	43.4		
		MCWB	37.5	35.5	42.6	47.2	53.9	59.9	62.5	60.7	57.3	50.0	43.1	36.6		
	10%	DB	40.4	38.0	49.2	59.2	70.0	75.2	82.9	82.9	74.5	59.7	49.1	39.1		
		MCWB	35.1	33.3	39.7	44.3	52.1	57.6	61.3	59.9	55.2	47.1	40.3	34.1		
	0.4%	WB	42.5	43.4	48.6	54.2	59.3	66.6	69.4	66.2	62.9	57.3	49.8	43.5		
		MCDB	50.5	52.6	62.3	73.2	78.8	77.8	85.9	82.2	80.7	74.4	64.4	51.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.4	38.0	45.8	50.7	57.4	63.4	66.6	64.4	60.5	53.9	46.6	39.8		
		MCDB	45.9	44.5	58.2	67.6	75.6	79.4	81.3	80.6	81.5	66.6	57.3	46.9		
	5%	WB	37.5	35.8	43.0	48.0	55.6	61.2	64.6	62.8	58.3	50.7	43.5	36.6		
		MCDB	43.5	41.4	53.5	62.2	71.6	75.8	79.3	81.3	76.9	62.1	52.8	43.0		
	10%	WB	35.3	33.5	40.1	45.1	53.6	59.1	63.0	61.3	56.3	48.1	40.6	34.0		
		MCDB	40.1	37.7	48.8	57.2	66.8	72.5	78.4	78.7	73.3	59.0	48.5	39.1		
	5% DB	MDBR	18.2	19.2	19.9	22.1	23.1	22.4	25.9	25.5	24.0	21.5	18.7	17.6		
		MCDBR	17.9	21.6	25.4	30.3	30.1	28.8	31.9	32.3	32.7	27.9	23.2	18.6		
		MCWBR	13.0	15.6	14.3	15.1	13.3	11.9	11.7	11.4	12.9	13.9	13.2	12.4		
		MCWBR	17.9	20.9	24.8	28.3	26.3	25.9	26.6	27.5	30.4	25.5	22.8	19.0		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.263	0.297	0.353	0.376	0.365	0.368	0.385	0.338	0.299	0.267	0.251		
		taud	2.352	2.471	2.441	2.287	2.268	2.361	2.366	2.290	2.400	2.490	2.481	2.417		
	Ebn at Noon		246	286	291	282	279	283	280	269	272	266	252	242		
		Edh at Noon	21	23	29	38	41	37	36	37	30	23	18	16		
(44) All-Sky Solar Radiation	RadAvg		395	711	1135	1562	1855	2014	2190	1768	1267	786	439	307		
	RadStd		38	59	78	108	129	122	114	116	105	80	33	27		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page